

Nitish Prasad

Manager, Data Science – AI and Deep Learning Solutions

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Open to Relocate

Professional Summary

MBA and Engineering graduate with 12 years of multi-domain experience across Automotive, Supply Chain, Logistics, Fintech, Banking, and Mortgage Analytics, currently serving as Manager, Data Science at Michelin India Pvt Ltd. Proven track record in architecting, building, and deploying deep learning models, feature engineering pipelines, embedding-based retrieval systems, and agentic ML workflows in high-throughput production environments. Technical expertise spans deep learning frameworks (PyTorch, TensorFlow), prompt-driven experiment orchestration (LangGraph, LangChain), vector databases, and enterprise cloud ML infrastructure on Azure Databricks and AMLS. Adept at driving autonomous experiment generation, aligning offline evaluation metrics to measurable business ROI, and leading cross-functional engineering teams.

Core Technical and Strategic Skills

- **Deep Learning, ML & Architecture:** PyTorch, TensorFlow, Deep Learning Time-Series (NBEATS, TFT), Feature Engineering, Embedding Models, Vector Databases, Predictive Modeling, Offline Model Evaluation & Metrics
- **Agentic ML, AI Research & NLP:** Agentic AI Workflows, Prompt-Driven Orchestration, Autonomous Experiment Pipelines, LangGraph, LangChain, RAG Architectures, LLMs, Prompt Engineering, SpaCy, Hugging Face, Aspect-Based Sentiment Analysis
- **Cloud, MLOps & Production Reliability:** Azure Databricks, Azure Machine Learning Services (AMLs), Scalable Backend Microservices, Production ML Pipelines, High-Throughput Serving, Model Drift & Health Monitoring, Python, SQL, Streamlit, Power BI
- **Domain Expertise:** Global Consumer Analytics, Supply Chain and Logistics, Automotive, Financial Services, Banking, Mortgage Analytics, Business Workflows
- **Leadership & Strategy:** AI Solution Architecture, Technical Mentorship, Cross-Functional Engineering Collaboration, Product Roadmapping, Technical Storytelling

Professional Experience

Manager, Data Scientist

Michelin India Pvt Ltd, Pune, India | Sep 2022 – Present

- **Directed** the design and production deployment of an autonomous forecasting agent using LangChain and LangGraph, establishing prompt-driven experiment pipelines (data profiling, imputation, model selection, explainability) that reduced manual planning cycle time by 40 percent.
- **Orchestrated** cross-functional deployment of deep learning models (NBEATS, TFT) built on PyTorch and TensorFlow for high-throughput operational workloads, improving forecast accuracy by 10 percent MAPE and delivering EUR 300K in value.
- **Managed** production ML pipeline deployments and backend microservices on Azure Databricks and AMLS, establishing continuous monitoring for model health, feature pipeline integrity, and low-latency performance.
- **Architected** an enterprise decision engine using vector databases, embedding models, and RAG architectures to automate feature analysis and sentiment synthesis, cutting analysis turnaround time by 35 percent.
- **Led** agentic ML research initiatives using prompt engineering, Azure OpenAI, and LLM workflows to autonomously generate feature extraction and text analytics experiments across complex domain datasets.
- **Engineered** a conversational data interface utilizing Databricks Genie and SQL, enabling direct natural language query translation and rapid feature extraction across enterprise datasets.
- **Delivered** aspect-based sentiment analysis NLP models using Python and SpaCy on customer feedback streams, driving operational updates that increased Customer Satisfaction (CSAT) by 8 percent.
- **Mentored** junior and mid-level data scientists on deep learning best practices, experiment tracking, code quality, and emerging agentic development workflows.

Data Scientist

Cognizant Technology Solutions (Client: ING Bank), Pune, India | Oct 2021 – Sep 2022

- **Fine-tuned** custom NLP models using SpaCy and Hugging Face frameworks for intent recognition and entity extraction, increasing overall prediction accuracy by 14 percent across production datasets.
- **Led** enterprise conversational AI system upgrades from Dialogflow ES to CX, expanding intent recognition scalability and reducing support escalations by 18 percent.
- **Established** model error analysis frameworks using statistical evaluation and confusion matrices to systematically identify signal patterns, accelerating offline model optimization cycles.

Analyst

AkzoNobel India Ltd, Pune, India | May 2020 – Oct 2021

- **Developed** predictive ML models in direct consultation with finance teams to forecast invoice payment timeliness, reducing overdue payment delays by 15 percent and improving cash flow predictability.
- **Formulated** logistics KPI predictive models to optimize supply chain operations and improve delivery SLA compliance by 12 percent.
- **Designed** enterprise Power BI dashboards with Row Level Security (RLS) for secure, automated executive reporting across global business units.

Data Analyst

A.P. Moller - Maersk, Pune, India | Jul 2017 – May 2020

- **Engineered** financial time-series forecasting models (achieving approximately 84 percent accuracy) using statistical methods, improving planning confidence for finance stakeholders.
- **Developed** ML customer risk clustering models for financial strategies, supporting more targeted risk-based decision-making.

Financial Analyst

XL Dynamics (Client: Sun West Mortgage Company), Mumbai, India | May 2014 – Apr 2017

- **Applied** statistical trend modeling and descriptive analytics on retail banking portfolios to evaluate credit performance metrics and default distributions.
- **Engineered** statistical anomaly and outlier detection frameworks on loan application datasets to identify duplicate entries and reduce operational risk across banking ERP systems.

Education

Master of Business Administration (MBA), Financial Management

University of Pune | 2012 – 2014

Bachelor of Engineering (BE), Electronics and Telecommunications

University of Pune | 2008 – 2012

Certifications and Awards

- **Certifications:** Complete Agentic AI Bootcamp with LangGraph and LangChain; Python for Time Series Analysis; Master Statistics and Machine Learning; SQL and Database Design A-Z
- **Key Awards:** Spot Award, Importer Sellout Forecasting, Michelin India Pvt Ltd, 2025; Team Award, Logistics KPI Forecasting, AkzoNobel India Ltd, 2021; Team Award, Overdue Report Automation, AkzoNobel India Ltd, 2021; Champ of the Month, Customer Risk Classification, A.P. Moller - Maersk, 2019; Champ of the Month, Working Capital Prediction, A.P. Moller - Maersk, 2019